

generate_samples_tsv.py

Overview

`generate\_samples\_tsv.py` scans a list of FASTQ file paths, detects paired-end reads (R1 and R2), and generates a **samples.tsv** file. The output is suitable for bioinformatics workflows (e.g., Snakemake pipelines) that require a mapping of sample IDs to their respective FASTQ files.

Features

- Reads a text file containing relative FASTQ file paths.
- Automatically converts relative paths to absolute paths.
- Groups R1 and R2 FASTQ files by sample prefix.
- Ensures only valid paired-end samples are included in the output.
- Saves results as a tab-separated values (TSV) file.

Input Requirements

1. FASTQ file list (`sorted\_fastq\_files.txt`):
 - Contains relative paths to `.fastq.gz` files.
 - Each line should point to a FASTQ file.
 - Example:
NG-13640/NG-13640_R1.fastq.gz
NG-13640/NG-13640_R2.fastq.gz
NG-29643/NG-29643_R1.fastq.gz
NG-29643/NG-29643_R2.fastq.gz
2. Supported FASTQ file naming conventions:
 - `\_R1.fastq.gz` / `\_R2.fastq.gz`
 - `\_1.fastq.gz` / `\_2.fastq.gz`

Output

- File: `config/samples.tsv`
- Format: Tab-separated file with columns:

sample	r1	r2
NG-13640	/abs/path_R1.fastq.gz	/abs/path_R2.fastq.gz

Usage

```
python3 generate_samples_tsv.py
```

Configuration

Inside the script:

- `data\_dir`: Root directory containing all FASTQ files.
- `txt\_path`: Path to `sorted\_fastq\_files.txt`.

Example:

```
data_dir = "/home/user/data/gatc"  
txt_path = "/home/user/scripts/sorted_fastq_files.txt"
```

Example Workflow Integration

```
SAMPLES = pd.read_csv("config/samples.tsv", sep="\t").set_index("sample").T.to_dict()
```

Dependencies

```
Python 3.6+  
pandas  
pip install pandas
```

Author

Generated for DSMZ sequencing pipeline automation.